$$\left[\underline{u_x w_x} + u w_{xx} + \underline{\underline{u_y w_y}} + u w_{yy} + \underset{\sim}{u_z w_z} + u w_{zz}\right] -$$

$$-\left\{\underline{u_x w_x} + \underline{\underline{u_y w_y}} + \underset{\sim}{u_z w_z}\right\} = u w_{xx} + u w_{yy} + u w_{zz} \equiv LHS \checkmark$$

hence verifies the given relation. ▨

Prob.2. (2&T, Ch.6, 1.2 modif-d) (?0) why do we need it?

Let u be in $C^2(\Omega)$ & and in $C^1(\Omega)$ where Ω is a "normal bounded domain" (means regular dom-n, i.e. Lipshitz, bounded, domain). Suppose that

(1) $$\underline{\nabla}^2 u = 0 \qquad \Omega$$

(2) $$u = 0 \qquad S,$$

where $S = \partial\Omega$, the boundary of the dom-n. (i) Show that $u = 0$ in Ω. (ii) Explain what this means in terms of the steady state heat eq-n problem?

Hint: Assume $\underline{x} \in \mathbb{R}^3$ & note that this argument can easily be extended to $\mathbb{R}^n$. Use one of the Green's identities with $w = u$ & use the fact that if the integral of a cont-s nonnegative f-n over Ω is zero, the the f-n is identically zero in Ω.

Soln: [Preamble]

(i) By hint, the fact:

(3) $$\int_\Omega f(\underline{x})\, d\underline{x} = 0 \quad \text{for } f(\underline{x}) \geq 0, \ \& \ f(\underline{x}) \in C^0(\Omega) \Rightarrow$$

$$f(\underline{x}) \equiv 0.$$

Recall Green's first identity: outward flux surface integral, over tiny dS

(4) $$\int_\Omega u \cdot \underline{\nabla}^2 w \, d\underline{x} = \int_{\partial\Omega} u \frac{\partial w}{\partial \underline{n}} dS - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w \, d\underline{x}$$

(ii) Explaining steady state sol-n:

We have $\boxed{u=0} \Rightarrow \boxed{\nabla u = \underline{0}}$ & of course $\nabla^2 u = 0$,

for again $\forall \underline{x} \in \mathbb{R}^3$,

$$(9) \quad \begin{cases} u_{xx} + u_{yy} + u_{zz} = 0 \\ u_x = u_y = u_z = 0 \\ u(x,y,z) = 0 \end{cases}$$

which is a steady-state $\Rightarrow$ objects' temperature is always zero ($u=0$) & it is not changing over time ($\nabla u = 0$).

$\blacksquare$

Prob. 3. (ZT, Ch 6, 1.3 (mod-d)). ?2 why needed?

Let u be in $C^2(\Omega) \cap C^1(\bar{\Omega})$, where Ω is a regular domain in $\mathbb{R}^n$. Suppose that

$$(1) \quad \nabla^2 u = 0 \qquad \Omega$$

$$(2) \quad \frac{\partial u}{\partial \underline{n}} = 0 \qquad \partial\Omega$$

WTS that (i) is a const. in $\bar{\Omega}$. The text says Ω is a "normal bounded domain" - this is the same as being "regular". Assume $\underline{x} \in \mathbb{R}^3$ & notes are the same with previous Prob. 2, (ii) Explain steady state.

Sol-n. [Preamble].

Let $\underline{x} \in \mathbb{R}^3$ & the following calculations hold for $\underline{x} \in \mathbb{R}^n$, WLOG.